Jinglan (Bella) DAI
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EDUCATION

Washington University in St. Louis, St. Louis, MO 08/2024 - 07/2028
Bachelor of Arts, Major: Statistics and Economics, Minor: Classics GPA: 3.98/4.00

- Coursework: Machine Learning, Bayesian Statistics, Linear Algebra, Probability, Linear Regression, Econometrics, Principles of Microeconomics, Principles of Macroeconomics, Latin
- Fall 2024, Spring 2025, & Fall 2025 – Dean's List in the College of Arts & Sciences

RESEARCH INTERESTS

Conformal Prediction in Hierarchical Data; Machine Learning; Robust High-dimensional Statistical Estimation and Inference; Biomedical Informatics

RESEARCH EXPERIENCE

Undergraduate Independent Study: *2-layer Hierarchical Time-to-event Conformal Prediction*, St. Louis, MO 01/2026 – Present

- Reviewed and summarized the algorithms in the related literature about 2-layer hierarchical data and time-to-event conformal prediction;
- Developed theoretical guarantees and algorithms on the 2-layer hierarchical time-to-event conformal prediction;
- In progress of developing a robust conditional Lower Prediction Bound (LPB) of data in hierarchical structures.

Research Assistant at WashU Cox's Lab: Duckweed Biology, St. Louis, MO 01/2026 – Present

- Formulated and standardized growth media for *Wolffia australiana* to ensure reproducible experimental conditions;
- Acquired high-resolution microscopic images of chlorophyll and performed quantitative image-based analysis using computational workflows; Extracted, processed, and analyzed chlorophyll-related measurements to support statistical evaluation of plant growth and physiological variation;
- Maintained experimental reproducibility through systematic solution preparation, protocol adherence, and laboratory quality control procedures.

Machine Learning: *High-Dimensional Genomic Analysis for Leukemia Classification*, St. Louis, MO 11/2025 – 01/2026

- Analyzed gene expression profiles from 72 leukemia patients to develop predictive models for subtype classification;
- Applied parametric (logistic regression, naive Bayes, discriminant analysis) and non-parametric approaches (classification trees, random forests, boosting, K-NN, SVM);
- Implemented tests to avoid overfitting, including model validation and feature filtering, to ensure classification accuracy.

Social Research on MBTI Heat, *Initiator & Team Leader*, Wuhan, China 11/2022 – 05/2023
China Thinks Big Project, Qualified for 2023 Global Final

- Convened and coordinated a team of five to study the phenomenon of peers relying on MBTI test results;
- Collected 118 questionnaires, revealing a lack of objectivity and standardization in the test;
- Conducted literature analysis to examine the physiological, psychological, and market drivers of adolescents' need for belonging;
- Synthesized findings into a research paper integrating theoretical and experimental evidence.
- Created a 5-minute interactive video, presenting the scientific validity of the MBTI and preventive programs for specific situations.

Statistical Modelling: *CO_2 Emission and Global Temperature Change*, *Member.* 11/2022 – 02/2023

- Use Python to fit the data of the atmosphere CO_2 concentration over the past 50 years into the quadratic polynomial regression model and quadratic polynomial+dummy variables model, and predict the CO_2 concentration by 2050 and 2100; introduce the ARIMA model to predict land-ocean temperature changes.
- Conduct correlation and regression analysis to determine the relationship between CO_2 concentration and land-ocean temperature changes.

PROFESSIONAL EXPERIENCE

WashU Residential Advisor, St. Louis, MO 08/2026 – 05/2027

- Built and maintained an inclusive, safe residential community by supporting residents' academic, social, and personal well-being;
- Enforced housing policies and served on on-call rotations, responding to incidents and coordinating with professional staff as needed;
- Planned and led community-building programs while connecting residents to appropriate campus resources.

WashU Department of Statistics and Data Science, St. Louis, MO 08/2025 – 12/2025

- *Grader* for [SDS 4010] Probability at Washington University in St. Louis
- *Statistics Help Room Tutor* at Washington University in St. Louis

WashU Arts & Sciences Pre-College Program, *Program Assistant,* St. Louis, MO 04/2025 – 07/2025

- Help to implement academic, community, and college readiness activities, provide information about university resources, provide academic and community support, inform students about policies and procedures, and enforce these policies and procedures.

WashU Office for International Student Engagement, *WU GUIDE,* St. Louis, MO 04/2025 – 08/2025

- Guide and support new international students through their transition by providing mentorship, campus tours, and support for academic and social integration.

EXTRACURRICULAR EXPERIENCE

Adopt A Grandparent (AAG), *Treasurer*, St. Louis, MO 08/2024 - Present

- Weekly volunteer at two local retirement communities;
- Manage budgeting and funding expense records while enhancing financial skills through literacy training.

SKILLS AND INTERESTS

- **Languages**: Chinese Mandarin (Native), English (Proficient), Latin (Familiar), Korean (Beginner)
- **Technical Skills:** Python (Proficient), R (Proficient), SQL (Intermediate), Excel (Intermediate), JavaScript (Basic), LaTex (Basic).